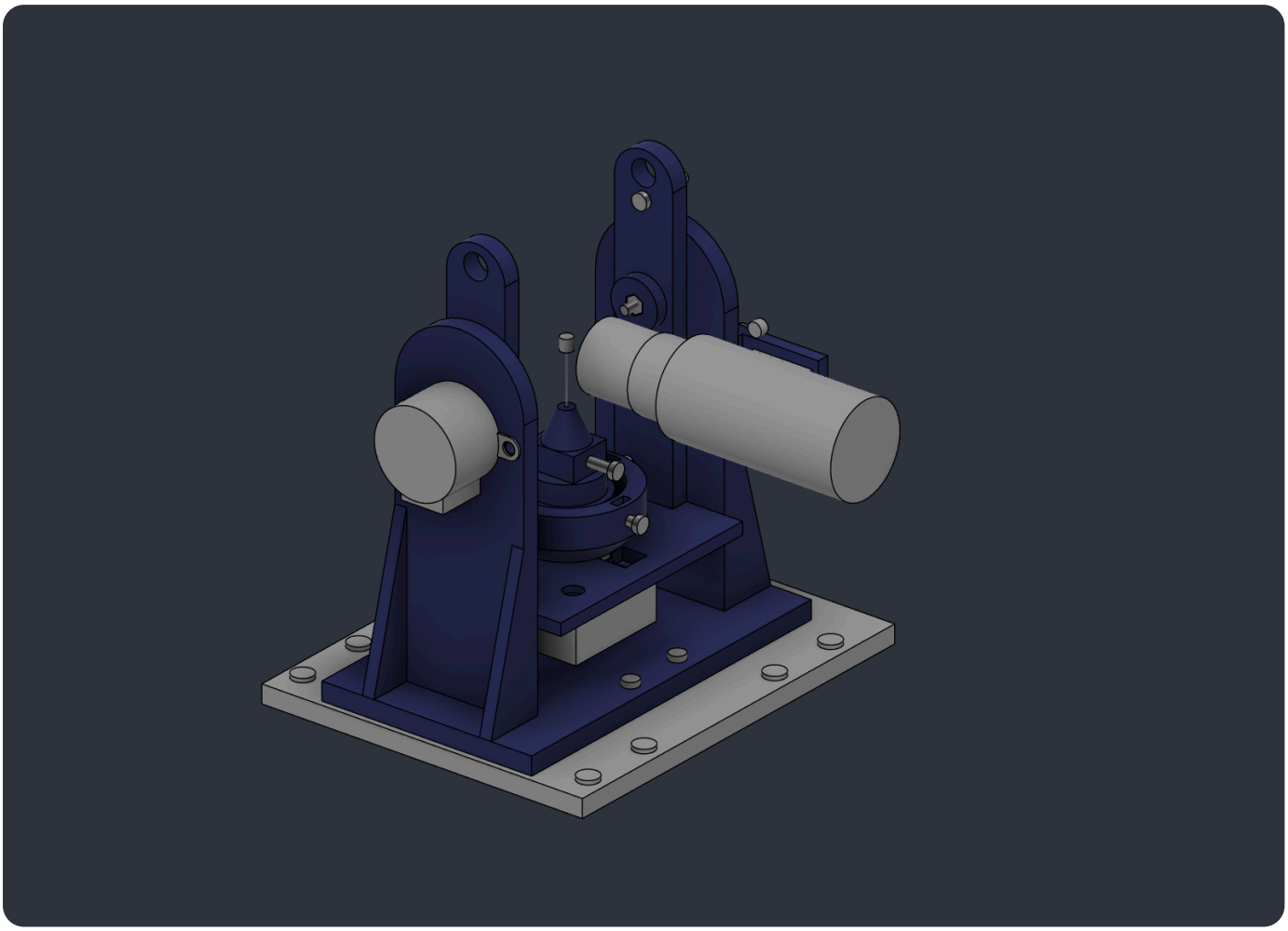
MacroRail scan module

Assembly and use · v1 · 2026-10-21



The module on the rail carriage, with the objective for scale.

Tilt — the cradle swings between the fork's uprights. A 28BYJ-48 drives it directly on the −Y side; the +Y side turns in a 608 bearing. Range −33° (belly towards the lens) to +90° (looking straight down on the back). Home: a microswitch.

Yaw — the head sits straight on the shaft of a second 28BYJ-48 under the cradle floor and turns all the way round. Home: a magnet in the head passes a hall sensor.

Centring — the insect has to sit where the two axes cross. Two M3 screws push the centring plate in X and Y, a spring pushes it back. ±2.5 mm each way. Done once per specimen, with the app's wizard.

Specimen holder — one printed part: the centring plate with the clamp column and nose on top. It stays in the head; to change specimens you swap the pin.

1 · Parts

Printed (PLA, 0.2 mm layers, no supports)

Part	Qty	Face on the bed	Notes
Fork	1	Base underside	Bolts to the rail carriage; holds the tilt motor and the 608.
Cradle	1	Floor underside	Swings on the tilt axis; carries the yaw motor and the head.
Yaw Head	1	Bell underside	No supports. Snap off the thin fin under the spring housing.
Specimen Holder	1	Plate underside	Centring plate, clamp column and nose in one part. Nose up.
Pin Pad	1	Any flat face	
Spring Plunger	1	Spigot end	Tiny: use a brim.
Tilt Axle	1	Flange	Shank vertical, so it comes out round.
Axle Spacer	1	Flat	
Carriage Washer	4	Flat	
Height Gauge	1	Underside	A tool, not mounted.
Endstop Bracket	1	Back (flat side)	Holds the tilt home switch on the +Y upright.

The STLs are in model coordinates: use *Place on face* in the slicer. Files: `exports/final/`.

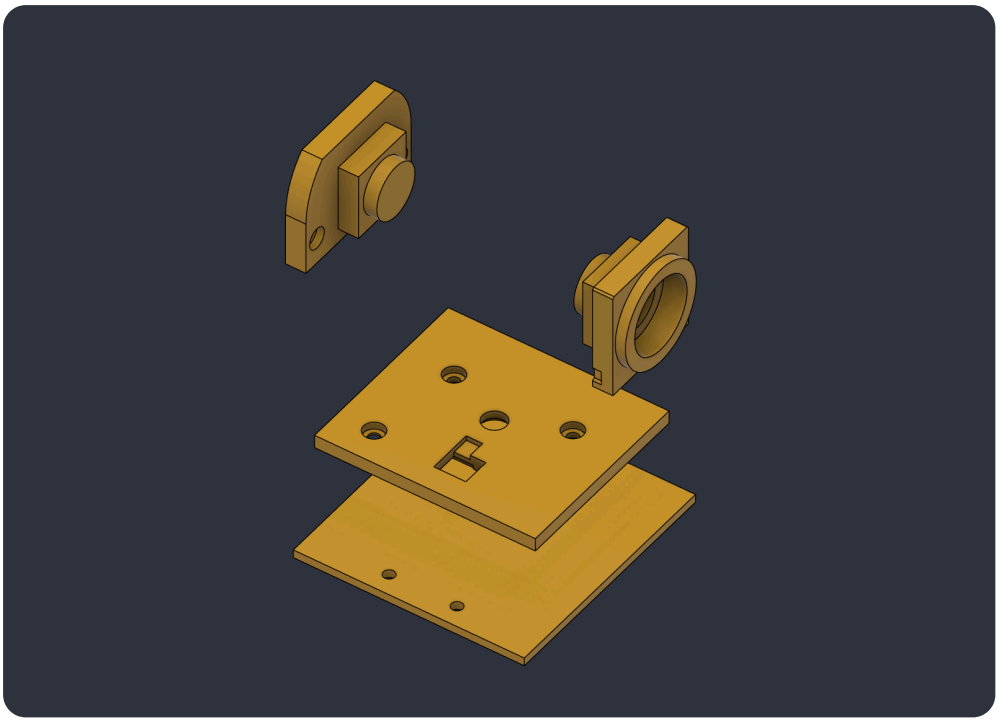
Hardware

Item	Qty	Where
28BYJ-48 + ULN2003 board	2	Tilt (outside the −Y upright), yaw (under the cradle floor)
608 bearing	1	+Y upright
KY-024 hall board	1	Under the cradle floor
Magnet Ø5.7 × 1.8	1	Hub underside (yaw home)
Pen spring Ø4.6 × 19	1	Spring housing, behind the plunger
M3 × 17.5 + nut	5	Base to carriage (4, with printed washers), tilt home flag (1)
M3 × 11.8 + nut	11	Tilt motor (2), yaw motor (2), KY-024 (1), centring screws (2), pad screw (1), set screw (1), endstop bracket (2)
M3 × 27.7 + nut	1	Tilt axle
Microswitch 20 × 10 × 5.8	1	Tilt home, on a printed bracket on the +Y upright. Held by two small Ø1.9 wood screws plus superglue.
Pin size 00, 0 or 1		Specimen pin

All screws are M3 button heads, turned with a **2 mm hex key**. Socket-head caps also fit (2.5 mm key). Lengths are under the head.

2 · Test pieces first

Print these before the big parts. Each is a thin slice of the real part, so a failed fit costs minutes, not hours. Files: `exports/test-pieces/`.

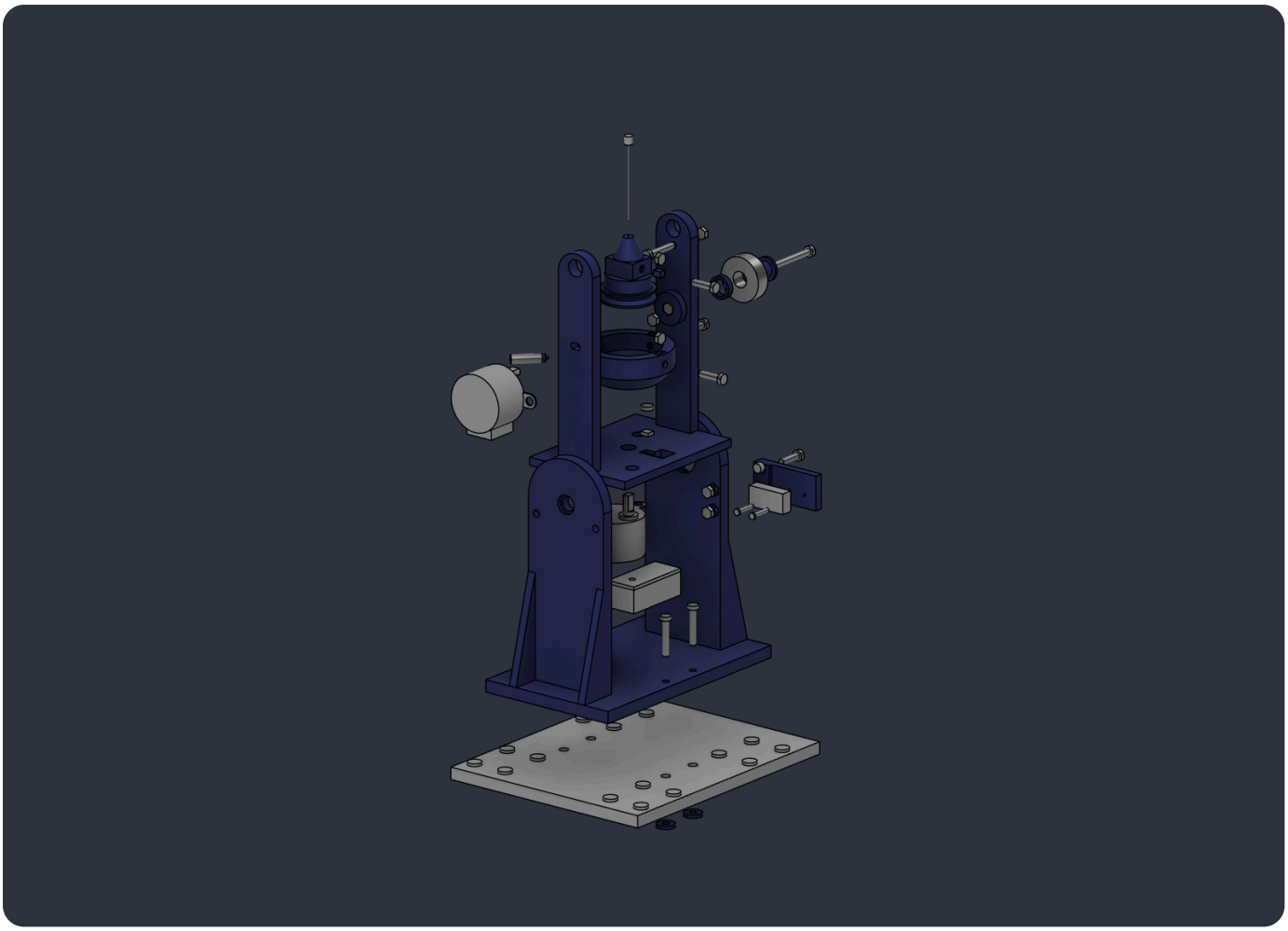


The six slices: motor seat and D-bore (left), bearing seat and axle socket (right), cradle floor and base holes.

Piece	It passes when
tp-bearing-seat	The 608 presses in (hand or vice) and doesn't fall out; the lip stops it.
tp-motor-seat	The tilt motor boss drops into the recess, both ear bolts line up, the counterbored heads sit flush.
tp-base-holes	A 2.5 mm slice of the fork base: lay it on the carriage, all four bolts line up, it lies flat over the carriage's screw heads.
tp-tilt-d-bore	Slides onto the motor shaft with no play and no force.
tp-axle-socket	The axle's spigot fits; the nut presses into its pocket.
tp-cradle-floor	Yaw motor boss and ear bolts, KY-024 bolt and board seat, sensor pocket and lead slot.
Yaw Head (whole part)	Fits the yaw shaft; nuts, magnet and spring go in; the fin snaps off cleanly.
Specimen Holder	Slides freely in the ring between the screw tips and the plunger; a pin passes through the nose hole.
Tilt Axle + Spacer	The axle slips through the 608; the spacer slides over it.
Endstop Bracket	Bolts to the upright; the switch's holes line up with the pilot holes; the wood screws bite.
Pin Pad + Plunger	The pad slides into its pocket and clamps pins 00 to 1; the plunger slides freely in its bore.

> If a fit is off, change the matching parameter in Fusion and re-export: `fit_press` (608 seat), `fit_D` (D-bores), `fit_slide`, `boss_clr`, `m3_pocketAF` (nut pockets), `magPocketD`, `pinHoleD`. Everything else follows.

3 · Assembly



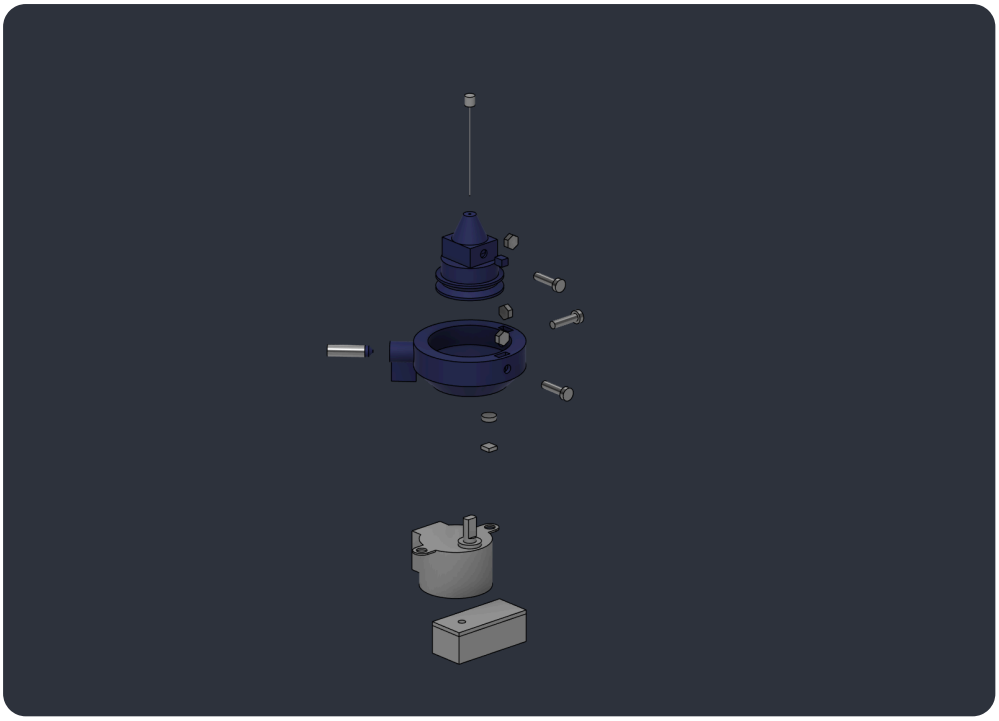
Everything in its place, pulled apart along the direction it goes on.

Yaw head

1. **Nuts.** Press an M3 nut into each slot in the ring wall (X and Y screws) and one into the hub slot (set screw).
2. **Hub magnet.** Glue one magnet into the pocket in the bell's underside. Any pole.
3. **Spring.** Snap off the fin under the spring housing. Drop the spring in, then the plunger, cone first.
4. **Centring screws.** Turn the X and Y screws in from outside until they just show inside.
5. **Holder.** Slide the pin pad into the holder's side pocket and the pad nut into its slot, and turn the pad screw in until it just touches. Lay the holder in the ring, its rim between the two screw tips and the plunger.

Cradle

1. **KY-024.** Bend the hall sensor (the small black TO-92) flat, sensing face up. Bolt the board under the floor with one M3 × 11.8; the sensor lies in the floor pocket, its legs in the slot.
2. **Yaw motor.** Push its shaft up through the floor and bolt the two ears (M3 × 11.8, heads in the counterbores on top).
3. **Head.** Slide the head onto the shaft, the D-flat to the set screw. Leave the 1 mm gap to the floor and tighten the set screw.



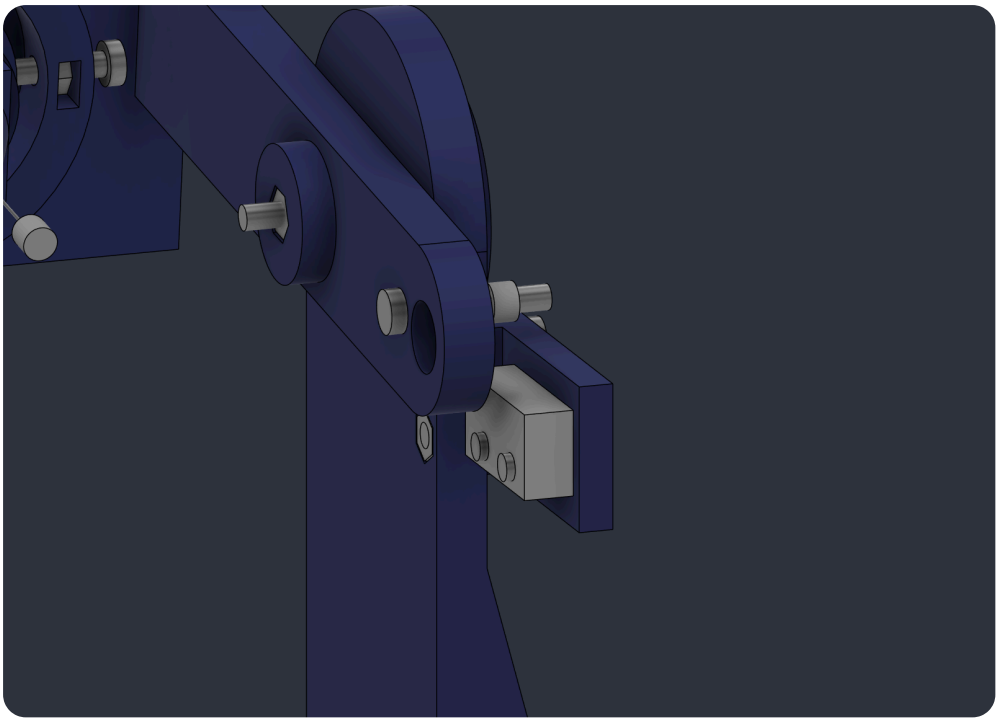
Yaw stack, top to bottom: pin, specimen holder with pad, pad screw and nut, head with screws, nuts and plunger, hub magnet, hall sensor, yaw motor, KY-024.

Fork and tilt

1. **Bearing.** Press the 608 into the +Y upright from the inside until it hits the lip.
2. **Tilt motor.** Fit it on the outside of the −Y upright, boss in the recess. Bolt the ears with two M3 × 11.8 from the inside (heads in the counterbores), nuts on the ears.
3. **Cradle.** Lower the cradle between the uprights and push the −Y hub onto the tilt shaft.
4. **Axle.** Spacer between the cradle and the bearing; push the axle through the 608 from outside into the +Y hub. M3 × 27.7 through the axle into the nut in the hub. Snug, not tight: the cradle must swing freely.
5. **Flag.** M3 × 17.5 through the hole near the top of the +Y arm, head inside, nut outside. This is what presses the home switch.
6. **Endstop.** Screw the switch onto the bracket (pivot end towards the camera, pins down) with the two wood screws and a drop of superglue. Press two M3 nuts into the pockets on the inside of the +Y upright and bolt the bracket to its outside with two M3 × 11.8.
7. **Carriage.** Set the fork on the carriage over its screw heads. Four M3 × 17.5 down through base and carriage, a printed washer and a nut underneath.

> Bolt stack under the carriage: base 6 + carriage 6 + washer 1.5 + nut 2.2 = 15.7 mm, so the 17.5 mm bolt shows 1.8 mm past the nut. The nuts sit at X ±30, Y ±10, clear of the lead nut, the bearing blocks and the screw.

4 · Homing



Tilt at +97°: the flag bolt on the +Y arm presses the roller of the switch on its bracket.

Tilt: a roller microswitch

The switch sits on a printed bracket bolted to the outside of the +Y upright with two M3 × 11.8 (nuts in the pockets on the inside). It hangs just past the upright's front edge, lever up, roller under the path of the flag: the M3 bolt through the +Y arm. Homing tilts the cradle slowly *away* from the lens, past +90°. The flag meets the roller at about +90° and the switch clicks at +97°; that position is the zero for every tilt after it. Homing away from the lens means a missed switch can never drive the insect into the objective.

Mount the switch with its pivot end towards the camera side and the pins down. Two small wood screws go through its holes into the pilot holes; a drop of superglue behind the switch keeps it from creeping. The click angle is the `home_t` parameter; the bracket follows it.

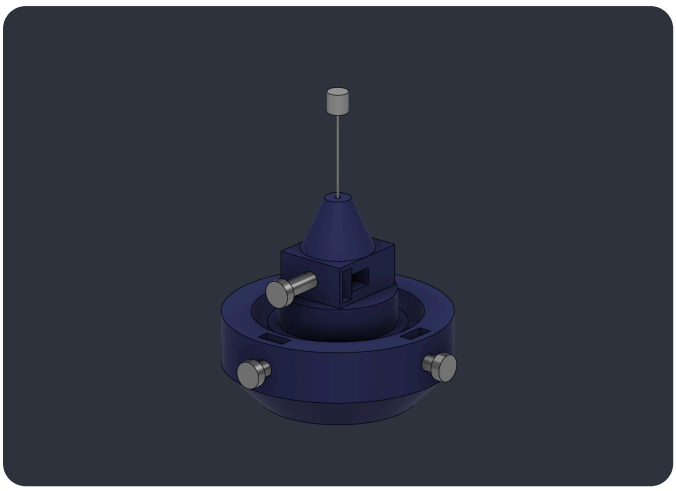
Yaw: magnet and hall sensor

The magnet in the head's hub passes 0.8 mm over the hall sensor once per turn. Homing turns the head until the KY-024's digital output switches; that is yaw zero. Set its threshold with the board's potentiometer so the LED lights only when the magnet is right above.

The holes at the top of the arms

The Ø8.3 holes 38 mm above the axis are for an optional counterweight. The cradle hangs below the axis, so tilting it to 90° pulls with about 40 mN·m. The 28BYJ-48's gearbox holds that, but if it ever skips steps, push an 8 mm rod through both holes and add nuts until the cradle balances.

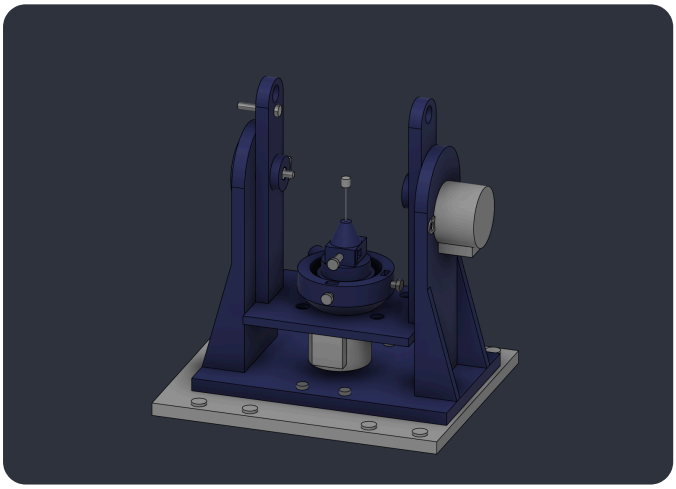
5 · Mounting a specimen



Specimen holder in the head: pin through the nose, pad screw clamping it from the side.

1. **Loosen.** Back the pad screw off a turn (hex key from the +X side; turn the yaw if it's out of reach). The holder stays in the head.
2. **Pin.** Push the specimen's pin down through the hole in the nose tip.
3. **Height.** Stand the height gauge on the nose tip, the pin in its slot. Slide the pin until the *middle of the insect* is level with the 19 mm step: that is the tilt axis height. The 17, 18, 20 and 21 steps show how far off you are.
4. **Clamp.** Tighten the pad screw: the nut is captive inside the holder, the pad presses the pin against the wall.

6 · Centring

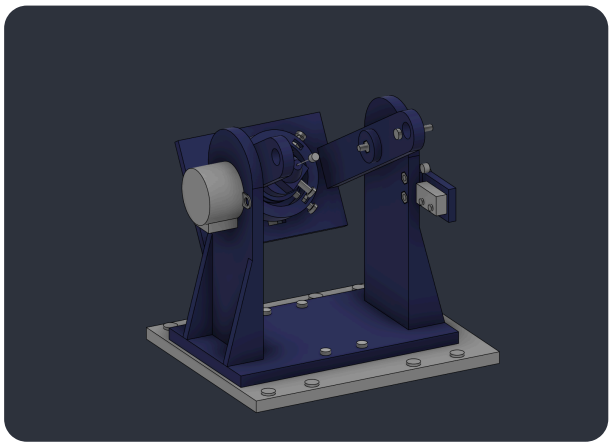


Yaw 180°: the screws point away from the lens, where the hex key reaches them.

The app's wizard shoots the insect at four yaw angles, fits a circle to where it drifts, and tells you how many turns to give each screw. It then turns the head so the screw faces you.

An M3 screw moves 0.5 mm per turn, so an eighth of a turn with the hex key is about 60 µm. That's finer than the wizard needs. Turn a screw in to push the plate away from it; the spring follows when you back it out. ±2.5 mm of travel covers any pin that's reasonably straight.

Centre once per specimen: every new pin puts the insect somewhere slightly different.



Tilt +60°, yaw 40°: the head turns towards the lens. Limits: −33° to +90°.